

OLABISI ONABANJO UNIVERSITY

Ago-Iwoye

Post UTME Pr Questions and Answers

MATHEMATICS



Compiled

by

CLASSIC EDUCATIONAL CONSULTS

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TIPS TO PASS POST UTME EXAMS

Do you want to pass post UTME of any tertiary institution in Nigeria? If yes, then you are in the right place. In this write up I will teach you best tips on how to prepare for Post UTME of any university or polytechnic in Nigeria. I enjoin you to read thoroughly, as I will highlight the ultimate preparation tips to pass post UTME screening examinations.

- **Have the mindset of a Success:**

Why is this, the first step in today's guideline? Having the mindset of a Successful student is first and surest step towards any form of progress in life. Be motivated to pass your Post UTME this year. If you don't continually prepare yourself and mind to pass this exam, there is nothing someone else can do to prepare you. Don't take what comes, go for what you want. Desire a Post UTME score and work towards making it a reality. Always tell yourself, "I must pass this Post UTME examination once and for all".

- **Don't get discouraged by your JAMB Score:**

Many a times, one of the main reasons why a lot of candidates fail to make it through the Post UTME exams is as a result of low self esteem due to their JAMB scores. One thing is to Score high marks in your JAMB Examination, another is to score high and beat the School's Departmental cut off marks. However, you need to develop a positive attitude toward your Post UTME irrespective of your Jamb score. Always strive to pass Post UTME examination irrespective of what you think you scored in Jamb. Post UTME is a very important factor that determines who gains admission. Even if you scored low in Jamb, you still have a golden opportunity to study your desired course at the University. A very high Post UTME score can help boost your Jamb result. One of the greatest mistakes candidates make in post utme is thinking that their high Jamb score is enough. Don't be deceived, no matter how high your Jamb score may be, a low Post UTME score will disgrace it.

- **Gather Relevant Information:**

A lot of candidates have insufficient or poor information about how the exam will play out, they have no clue whatsoever of how many questions they will be asked, what form the examination will be set in; and it is for these reasons that having relevant information is paramount.

You should know beforehand whether you are writing a paper based test or a computer based examination, your information gathering will help you know how much time is allotted to your paper, and will help you while studying to time yourself as you practice questions, giving you an



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edge over thousands of candidates who only find out about the time allotted for the paper only minutes before the exam commences. Also, you should check to know the terrain of the school. If possible, travel to the school and ask around. Ensure you have somewhere to stay a day or two before your examination (if out of state). This is important since you need to be well-rested before sitting for an examination if you hope to give it your best and perform well. Finally, try to gather as much information as you can about your university and department of choice, especially such things as required subject combination and cut-off mark from the previous year; as this could prove the difference between you and other inexperienced candidates, and whether you get an admission to your preferred course of study or not.

- **Avoid Procrastination:**

Later things... This is the language of procrastinators. They always think there is time. My dear, do whatever you need to do now. Time is an asset and not a handset. The later you always say doesn't come as expected. Learn to live your life with a sense of urgency. Have the mentality, "I am behind schedule...". Your Post UTME examination is not far. Say no to procrastination.

- **Study with Past-Questions:**

Dear friend, do not fall into the misguided conception that you can go it alone! It is important to have a clear understanding of the questions you are likely to be asked and the likely areas the questions might come from, and what better way to do that than with questions of previous examinations?

How do I get Post UTME past questions for my University of choice? It is very easy. Yes, it is! We have compiled FREE and RELIABLE post UTME past questions and answers for various Institutions. So, do try your best to get relevant past-questions that apply to your school and department, and go through it each day.

- **Set your Target:**

Your background check should tell you what the cut-off mark of your department of choice, hence your target point. You have to set a target for yourself and determine in your heart that you can achieve it, as you work towards it. Make sure the only thing that robs you of sleep is how to successfully meet your target score. Ensure that whatever target you set is above what you scored in



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UTME, since you will need to do better than your UTME performance to make the merit list of your department.

- **Reduce your time in Social Media:**

Some people live their entire life on Facebook, Twitter, Instagram, Whatsapp etc. This is so bad and disastrous if you are preparing for exams. Try and reduce your time spending on social media during this time. Maybe, after the exams you can go back and make it your second home, (just saying).

- **Learn To Remember:**

To pass Post UTME once and for all, you must learn to remember. It is not just enough to read. It is more important to remember what you have read. The difference between someone who reads for shorter hours but pass very well and another who studies for long hours but don't perform well is "remembrance". Study to remember and not to forget. If you can remember all you have read, then success will smile on you. Do whatever it takes to pass Post UTME examination genuinely.

- **Learn To Solve Without a Calculator:**

I know you cannot give me the correct answer to $2/3$ right now without a calculator. I can see that you are trying to get the answer. No, don't give me that look, I know just what I'm saying.

Most schools do not allow the use of calculator in their Post UTME examination. Therefore, you must learn to solve problems without pressing your calculator. It is time to temporarily break the covenant you have with calculator.

- **Give yourself enough time to rest:**

Another factor that causes low exam performance is STRESS! Stress as we all know it is very disastrous to the Human body. You need to give your body the maximum amount of rest it needs. Reading, reading and reading all the time isn't advisable at all, according to science, the Human brain is always at work. Give your brain time to relax from the previous tasks so as to recover and work efficiently.

NOTE: We can't cover everything in this article alone, we'll be continuing this discussion some other day.

Don't forget to share your thoughts with us @ classicconsults2005@gmail.com

Classic Educational Consults wishes you good Success in your examinations



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1. There are 8 green balls, 4 blue balls and 3 white balls in a box. Then 1 green and 1 blue balls are taken from the box and put away. What is the probability that a blue ball is selected at random from the box?

A. $\frac{3}{13}$ B. $\frac{2}{13}$ C. $\frac{4}{15}$ D. $\frac{3}{15}$

2. Find r, if $7r7_8 = 618_9$.

A. 3 B. 2 C. 6 D. 5

3. Simplify $(\frac{3}{4} \text{ of } \frac{4}{9} \div 9\frac{1}{2}) \div 1\frac{5}{19}$

4. A. $\frac{1}{5}$ B. $\frac{1}{4}$ C. $\frac{1}{36}$ D. $\frac{1}{25}$

A student measures a piece of rope and found it was $1.27m$ long. If the actual length of the rope was $1.25m$, what was the percentage error in the measurement?

A. 1.6% B. 1.0% C. 0.8% D. 0.16%

5. At what rate will the interest on N⁵⁰⁰ increase to N²⁵ in 5 years reckoning in simple interest?

A. 2% B. 1% C. 4% D. 5% If

6. $p:q = \frac{2}{3}:\frac{1}{6}$ and $q:r = \frac{3}{4}:\frac{1}{2}$ Find $p:q:r$

A. $12:3:2$ B. $12:15:4$ C. $9:10:15$ D. $9:12:15$ Evaluate

7. $(\frac{243}{32})^{\frac{-2}{3}} \times 2^{-2}$

A. 3 B. 6 C. $\frac{1}{6}$ D. $\frac{1}{3}$

8. Given that $\log 2 = 0.3010$, $\log 7 = 0.8451$, Evaluate $\log 224$

A. 2.1461 B. 2.3501 C. 2.0491 D. 3.1461

9. Rationalize $\frac{2\sqrt{5}+\sqrt{7}}{\sqrt{7}-\sqrt{5}}$

10. A. $\frac{3\sqrt{35}-\sqrt{17}}{2}$ B. $3\sqrt{35}+\sqrt{17}$ C. $3\sqrt{35}-\sqrt{17}$ D. $\frac{3\sqrt{35}+\sqrt{17}}{2}$ Express the product of 0.31 and 0.34 in standard form

A. 1.0541×10^{-1} B. 1.0541×10^{-2} C. 1.0541×10^{-3} D. 1.0541×10^{-4}



11. In a survey of 60 news paper readers, 49 read Nation and 30 read Punch, how many read both papers?

- A. 10 B. 5 C. 20 D. 15

12. Make R the subject of the formula if $P = \frac{M}{5}(X + R^2) + 2$.

- A. $\sqrt{\frac{5P+10+XM}{M}}$ B. $\sqrt{\frac{5P+10-XM}{M}}$ C. $\sqrt{\frac{5P-10-XM}{M}}$
D. $\sqrt{\frac{5P-10+XM}{M}}$

13. If $9x^2 + 6xy + 4y^2$ is a factor of $27x^3 - 8y^3$, find the other factor.

- A. $2y - 3x$ B. $2y + 3x$ C. $-2y - 3x$
D. $2y + 3x$

14. Factorize completely $\frac{x^2+2x^2-15x}{2x^2-18}$

- A. $\frac{x(x+5)}{2(x-3)}$ B. $\frac{x(x+5)}{2(x+3)}$ C. $\frac{x(x-5)}{2(x-3)}$ D. $\frac{x(x+5)}{2x+9}$

15. Solve for x and y if $x-y=3$ and $x^2 - y^2 = 9$

- A. (-3,0) B. (0,-3) C. (3,0) D. (0,3)

16. If y varies directly as the square root of x and $y=3$ when $x=25$. Calculate y when $x=100$.

- A. 12 B. 3 C. 5 D. 6

17. If x is inversely proportional to y and $x = 3\frac{1}{2}$ when $y=2$, find x if $y=4$.

- A. $1\frac{1}{4}$ B. $2\frac{3}{4}$ C. $1\frac{3}{4}$ D. $2\frac{1}{4}$

18. For what range of values of x is $\frac{1}{3}x + \frac{1}{4} > \frac{1}{4}x + \frac{1}{2}$?

- A. $x < 3$ B. $x > 3$ C. $x > -3$ D. $x < -3$

19. Solve the inequalities $-6 \leq 4 - 2x < 5 - x$.

- A. $-1 < x < 5$ B. $-1 \leq x \leq 6$ C. $-1 \leq x < 6$
D. $-1 < x \leq 5$

20. Find the sum to infinity of the following series $0.2 + 0.02 + 0.002 + \dots$

- A. $\frac{1}{4}$ B. $\frac{2}{9}$ C. $\frac{2}{11}$ D. $\frac{2}{7}$



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21. The 3rd term of an arithmetic progression is -8 and the 7th term is -28. Find the 10th term of the progression.
- A. -43 B. -164 C. 164 D. 44
22. If $x \times y = x - y^2$, find the value of $(2 \times 3) \times 5$
- A. -25 B. 25 C. -32 D. 32
23. If p and q are two non-zero numbers and $16(p + q) = (16 + p)/q$, which of the following must be true.
- A. $p < 1$ B. $p = 16$ C. $q < 1$ D. $q = 16$
24. If $\begin{vmatrix} x & 4 \\ 3 & 7 \end{vmatrix} = 9$, find the value of x.
- A. 4 B. 5 C. 2 D. 3
25. Evaluate $\begin{vmatrix} 3 & 0 & 6 \\ 5 & 7 & 4 \\ 9 & 0 & 2 \end{vmatrix}$
- A. -336 B. 336 C. 420 D. -420
26. A rectangular picture 6cm by 8cm is enclosed by a frame $(1/2)$ wide. Calculate the area of the frame.
- A. 15 sq cm B. 20 sq cm C. 13 sq cm D. 17 sq cm
27. The area of $3\frac{7}{8}$ and $1\frac{1}{3}$ is less than the difference between $\frac{3}{8}$ and $1\frac{2}{3}$ by
- A. $3\frac{11}{12}$ B. $5\frac{1}{4}$ C. $1\frac{1}{2}$ D. $8\frac{1}{8}$
28. Multiply $(x + 3y + 5)$ by $(2x^2 + 5y + 2)$
- A. $2x^3 + 3yx^2 + 10xy + 15y^2 + 13y + 10x^2 + 2x + 10$
- B. $2x^3 + 6yx^2 + 5xy + 15y^2 + 31y + 10x^2 + 2x + 10$
- C. $2x^3 + 3yx^2 + 5xy + 10y^2 + 13y + 5x^2 + 2x + 10$
- D. $2x^3 + 2yx^2 + 10xy + 10y^2 + 31y + 5x^2 + 2x + 10$
29. The sum of the progression $1 + x + x^2 + x^3 + \dots$ is equal
- A. $1/(1 - x)$ B. $1/(1 + x)$ C. $1/(x - 1)$ D. $1/x$
30. If $x^2 + 4 = 0$, then x =
- A. 4 B. -2 C. none of the se D. 2
31. Five years ago, a father was 3 times as old as his son. Now, their combined ages amount to 110 years. Thus, the present age of the father is



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- A. 75 years B. 60 years C. 98 years D. 81 years
32. If $y = 2x^2 + 9x - 35$, find the range of values for which $y < 0$.
- A. $-7 \leq x < 5$ B. $-5 \leq x < 7$ C. $-7/2 < x \leq 5$
- D. $-7 < x < 5 (5/2)$
33. Mother reduced the quantity of food bought for the family by 10% when she found that the cost of living had increased by 15%. Thus the fractional increase in the family food bill is now
- A. 1/12 B. 6/35 C. 19/300 D. 7/200
34. Given that $a * b = ab + b + a$ and $a^o b = 1 + b + a$. Find $(a * b)^o (a * c)$, if a, b, c are real numbers.
- A. $ac + ab + bc + b + c + 1$ B. $ac + ab + a + c + 2$
- C. $ac + ab + 2a + b + c + 1$ D. $ac + ab + b + c + b + c + 2$
35. If the four interior angles of a quadrilateral are $(P + 10)^\circ, (P - 30)^\circ, (2P - 45)^\circ$, and $(P + 35)^\circ$, then P is
- A. 78° B. 125° C. 135° D. 60°
36. Simplify $(a - b)/(a + b) - (a + b)/(a - b)$
- A. $4ab/(a^2 - b^2)$ B. $4ab/(a^2 - b^2)$
- C. $2ab/(a^2 - b^2)$ D. $2ab/(a^2 - b^2)$
37. The minimum point on the curve $y = x^2 - 6x + 5$ is at
- A. (1, 5) B. (3, -4) C. (2, 3) D. (3, 4)
38. If $3x - (\frac{1}{4}) > (\frac{1}{4}) - x$, then the interval of values of x is
- A. $x > (1/3)$ B. $x < (1/3)$
- C. $x < (9/16)$ D. $x > (9/16)$
39. A man runs a distance of 9km/h for the first 4km and then 2km/h for the rest of the distance. The whole run takes him one hour. His average speed for the first 4km is
- A. 6km/h B. 8km/h C. 9km/h D. 11km/h
40. In a soccer competition in one season, a club had scored the following goals: 2, 0, 3, 3, 2, 1, 4, 0, 0, 5, 1, 0, 2, 2, 1, 3, 1, 4, 1, and 1. The mean, median and mode are respectively.



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- A. 1, 1.8, and 1.5 B. 1.8, 1.5 and 1 C. 1.8, 1 and 1.5
D. 1.5, 1 and 1.8
41. If $\sec^2 \theta + \tan^2 \theta = 3$, the angle θ is equal to
A. 20° B. 60° C. 45° D. 90°
42. The set of values of x and y which satisfies the equations $x^2 - y - 1 = 0$ and $y - 2x + 2 = 0$ is
A. 1, 0 B. 1, 1 C. 2, 2 D. 0, 2
43. Two triangles have the same area if
A. two sides in one triangle are equal to two sides in the other.
B. three sides in one triangle are equal to three sides in the other.
C. two angles in one triangle are equal to two angles in the other.
D. three angles in one triangle are equal to three angles in the other.
44. If $25^{x-1} = 64(5/2)^6$, then x has the value
A. 7 B. 4 C. 32 D. 5
45. In a circle of radius 10cm, a chord of length 10cm is xcm from its centre. What is x.
A. $10\sqrt{2}$ B. $5\sqrt{2}$ C. $10\sqrt{3}$ D. $5\sqrt{2}$
46. The smallest number such that when it is divided by 8 has a remainder of 6 and when it is divided by 9 has a remainder of 7 is
A. 50 B. 70 C. 80 D. 60
47. Evaluate $\int_0^{\pi/4} \sec^2 \theta d\theta$.
A. $\frac{1}{4}$ B. $\frac{\pi}{2}$ C. 1 D. $\frac{\pi}{4}$
48. When $\frac{1}{4}$ dealer sells cycle for N81 he makes a profit of 8%. What did he pay for the bicycle?
A. N74 B. N74.52 C. N75 D. N75.52
49. Find the roots of the equation $10x^2 - 13x - 3 = 0$
A. $x = 3/5$ or $-1/2$ B. $x = -1/5$ or $3/2$ C. $x = 3/10$ or 1
D. $x = -3/10$ or 1
50. The median of the set of numbers; 4, 9, 4, 13, 7, 14, 10, 7 is



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A. 13 B. $\frac{7}{4}$ C. $\frac{7}{2}$ D. 10

51. List all the integer values of x satisfying the inequality $-1 < 2x - 5 \leq 5$.

A. 2, 3, 4, 5 B. 2, 5 C. 3, 4, 5 D. 2, 3, 4

52. The ratio of the areas of similar triangles is necessarily equal to

- A. the ratio of the corresponding sides.
B. the ratio of the square on corresponding sides.
C. the ratio of the corresponding heights of the triangles.
D. half the ratio of the corresponding heights of the triangles.

53. A man and his wife went to buy article costing ₦400. The woman had 10% of the cost and the man 40% of the remainder. How much did they have altogether.

A. ₦216 B. ₦200 C. ₦184 D. ₦144

54. Simplify $\log_{10} 8 / \log_{10} 4$

A. $\log_{10} 2$ B. $\log_8 4$ C. $\frac{3}{2}$ D. 2

55. Three numbers are connected by the relationship $y = 4x/9 + 1$ and $z = 4y/9 + 1$. If $x = 99$, find z .

A. $6\frac{1}{3}$ B. 20 C. 21 D. $176\frac{4}{9}$

56. In a school there are 35 students in class 2A and 40 in class 2B. The mean score for class 2A in a Mathematics examination is 60.00 and that for 2B in the paper is 52.5. Find, to one place of decimals, the mean of the combined classes.

A. 56.5 B. 56.0 C. 56.3 D. 56.2

57. A set of data contains a total of 130 items which are divided into six groups for display on a pie chart. If one of the groups contains 26 items then the sector representing this group on the pie chart contains an angle x° at the centre of the circle where x is

A. 3 B. 60 C. 70 D. 72

58. In triangle FGH, $\angle G = 90^\circ$, $\angle H = 60^\circ$, while triangle XYZ, $\angle X = 60^\circ$

$\angle Y = 30^\circ$

- A. $|YZ|/|ZX|$ B. $|YX|/|YZ|$ C. $|ZX|/|YZ|$
D. $|YZ|/|YX|$

59. A pentagon has four of its sides equal. If the size of the fifth angle is 60° find the size of each of the four equal angles.



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60. A. 60° B. 108° C. 120° D. 150°
61. The expression $(x^a/x^b)^{a+b} \cdot (a-b)^{a^2/b}$ is $x^{a^2-b^2}$
62. What will be the value of k so that the quadratic equation $kx^2 - 4x + 1 = 0$ has equal roots?
63. If it is given that $5^{x+1} + 5^x = 150$ then the value of x is equal to
64. Solve the system of equations $2^{x+y} = 32$ and $3^{y-x} = 27$
65. Simplify the given expression $\sqrt{\frac{1-\cos x}{1+\cos x}}$
66. Find the area of the curved surface of a cone whose base radius is 6cm and whose height is 8cm. (Take $\pi = 22/7$).
67. The expression $x^3 - 4x^2 + cx + d$ such that $x+1$ is its factor, and its value is 1 when x is -2. Find c and d.
68. If a function is defined by $f(x+1) = 3x^2 - x + 4$. Find f(0).
69. A cylindrical motor of height 12cm has uniform thickness of 2cm. If the diameter of its outer cross-section is 10cm, find the volume of the constituent material.



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6

- A. 9 cm B. $\sqrt{65} \text{ cm}$ C. $4\sqrt{2} \text{ cm}$ D. 7 cm
70. If x varies inversely as y , and y varies directly as the square root of z , and z varies directly as $\frac{1}{w^2}$, write down in words how x varies with w .
- A. x varies inversely as w^2 B. x varies directly as w^2
- C. x varies directly as w D. x varies inversely as w
71. Simplify $\frac{\sin^2 x}{(1 + \cos x)} + \frac{\sin^2 x}{(1 - \cos x)}$
- A. 2 B. $\sin x$ C. 1 D. $\sin^2 x$
72. From two points X and Y , 8cm apart, and in line with a pole, the angle of elevation of the top of the pole are 30° and 60° respectively. Find the height of the pole, assuming that X , Y and the foot of the pole are on the same horizontal plane and X and Y are on the same side of the pole.
- A. 4m B. $(8\sqrt{3})/3 \text{ m}$ C. $4\sqrt{3} \text{ m}$ D. $8\sqrt{3} \text{ m}$
73. A bag contains 3 apples, 4 oranges and 3 bananas. What is the probability of selecting a banana and then an apple?
- A. $9/100$ B. $9/10$ C. $1/10$ D. $2/3$
74. Evaluate ${}^nP_r - {}^{n-1}P_{r-1}$
- A. n B. $n-1$ C. $n-2$ D. $2n$
75. The chance of three independent events X , Y , Z occurring are $\frac{1}{2}$, $\frac{2}{3}$, $\frac{1}{4}$
- A. $\frac{1}{8}$ B. $\frac{1}{24}$ C. $\frac{1}{12}$ D. $\frac{1}{4}$
76. If $P = \begin{pmatrix} 2 & -1 \\ 3 & 3 \end{pmatrix}$, find P^{-1}
- A. $\begin{pmatrix} \frac{-1}{9} & \frac{-1}{9} \\ \frac{1}{9} & \frac{1}{9} \end{pmatrix}$ B. $\begin{pmatrix} \frac{1}{9} & \frac{1}{9} \\ \frac{-1}{9} & \frac{-1}{9} \end{pmatrix}$ C. $\begin{pmatrix} \frac{-1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{-1}{9} \end{pmatrix}$ D. $\begin{pmatrix} \frac{1}{9} & \frac{-1}{9} \\ \frac{-1}{9} & \frac{1}{9} \end{pmatrix}$
77. The interior angles of a quadrilateral are $(x + 20^\circ)$, $(2x - 45^\circ)$, $(x - 15^\circ)$ and $(2x + 10^\circ)$. Find the value of the least interior angle.
- A. 60° B. 70° C. 80° D. 90°
78. If the two smaller sides of right angled triangle are 8cm and 9cm, find its area
- A. 63 cm^2 B. 88 cm^2 C. 102 cm^2 D. 112 cm^2
79. An arc subtends an angle 60° at the centre of circle of radius 6cm. Calculate the area of the sector formed. ($\pi = \frac{22}{7}$)



$\frac{132}{7}$ B. $\frac{122}{7}$ C. $\frac{112}{7}$ D. $\frac{102}{7}$

80. A cylindrical pipe 40m long with radius 7m has one end open. What is the total surface area of the pipe?

A. 609π B. 658π C. 560π D. 98π

81. What is the locus of points equidistant from points P(1,4) and Q(2,5).

A. $y = -x - 6$ B. $y = x + 6$ C. $y = x - 6$ D. $y = -x + 6$

82. Find the distance between the points $(\frac{2}{3}, \frac{2}{3})$ and $(\frac{-1}{3}, \frac{-3}{3})$

A. 1 B. 0 C. $\sqrt{3}$ D. $\sqrt{2}$

83. Find the gradient of the line passing through the points p(1,2) and q(2,5)

A. 3 B. 2 C. 5 D. 4

84. Find the equation of a line perpendicular to $y = -4x + 2$ passing through (2,3)

A. $4y + x + 10 = 0$ B. $4y - x - 10 = 0$ C. $4y - x + 10 = 0$

D. $4y + x - 10 = 0$

85. If $\cot \theta = \frac{7}{15}$, where θ is acute, find $\tan \theta$.

A. $\frac{15}{8}$ B. $\frac{15}{7}$ C. $\frac{8}{17}$ D. $\frac{15}{17}$

86. If $y = (2x - 1)^3$, find $\frac{dy}{dx}$

A. $6(2x - 1)$ B. $3(2x - 1)$ C. $6(2x - 1)^2$ D. $3(2x - 1)^2$

87. If $y = x \cos x$, find $\frac{dy}{dx}$

A. $\sin x - x \cos x$ B. $\cos x - x \sin x$ C. $\cos x - \sin x$

D. $\sin x + \cos x$

88. At what value of x does the function $y = -3x + 2x + x^2$ attain a minimum value?

A. 1 B. -4 C. 4 D. -1

89. Evaluate $\int_0^2 (x^3 - x^2) dx$

A. $11\frac{3}{4}$ B. $12\frac{3}{4}$ C. $10\frac{3}{4}$ D. $11\frac{3}{4}$

90. Find $\int (\cos x + 2) dx$



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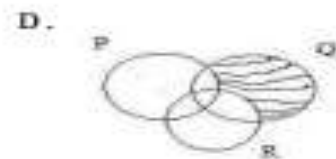
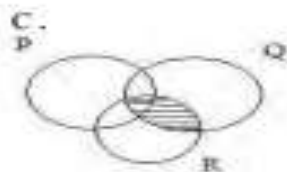
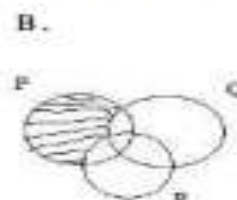
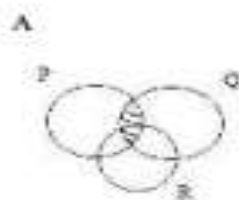
- A. $\sin x + 2x + k$ B. $-\sin x + 2x + k$ C. $\sin x + x^2 + k$
 D. $-\sin x + x^2 + k$

91.

Marks	2	3	4	5	6	7	8
No of Students	4	2	5	2	4	1	3

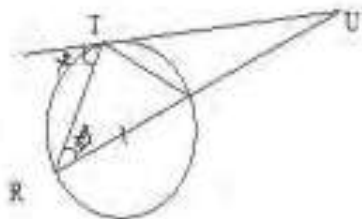
From the table above if the pass mark is 5, how many students failed the test?

- A. 7 B. 6 C. 11 D. 2
92. If three unbiased coins are tossed, find the probability that they are all tails
 A. $\frac{1}{6}$ B. $\frac{1}{3}$ C. $\frac{1}{9}$ D. $\frac{1}{8}$
93. In how many ways can a committee of 3 women and 4 men be chosen from 5 men and 5 women
 A. 250 B. 25 C. 50 D. 100
94. Find the standard deviation of 2, 4, 5 and 6
 A. $\sqrt{\frac{7}{2}}$ B. $\sqrt{\frac{2}{7}}$ C. $\sqrt{7}$ D. $\sqrt{14}$
95. Find the equation of a line parallel to $y = -3x + 2$ passing through (1, 3)
 A. $y + 3x - 6 = 0$ B. $y - 3x - 6 = 0$ C. $y - 3x + 6 = 0$
 D. $y + 3x + 6 = 0$
96. Which of the Venn diagrams below represents $P \cap Q' \cap R'$





97



From the diagram above, find x

- A. 55° B. 65° C. 50° D. 75°



98. From the cyclic quadrilateral PQRS above find the value of x .

- A. 30° B. 32° C. 60° D. 62°

99. If a and b are the roots of $x^2 - 5x + 7 = 0$, find $a^2 + b^2$

- A. 11 B. 25 C. -14 D. 39

100. Find, correct to three significant figures, the value of $\sqrt{41830}$

- A. 205 B. 647 C. 2050 D. 6470

101. Which of the following is not a factor of $12^4 - 5^4$?

- A. 169 B. 13 C. 17 D. 49

102. When a dealer sells a bicycle for ₦81, he makes a profit of 8%. What did he pay for the bicycle?

- A. ₦74 B. ₦76 C. ₦75.54 D. ₦75

103. The median of the set of numbers 4, 9, 4, 13, 7, 14, 10, 17 is

- A. 9.5 B. 7 C. 10 D. 8.5

104. List all the integer values of x satisfying the inequality $-1 < 2x - 5 \leq 5$



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A. 2,4,5 B. 1,4,5 C. 4,5,6 D. 3,4,5

105. A solid cylinder of radius 3cm has a total surface area of $36\pi\text{cm}^2$. Find its height.

A. 2cm B. 3cm C. 4cm D. 5cm

106. Simplify $\frac{\log 10^8}{\log 10^4}$

A. 1.5 B. 7 C. 3 D. 2

107. Write down the number 0.0052048 correct to three significant figures. 

A. 0.005 B. 0.0052 C. 0.00521 D. 0.00520

108. A man and his wife went to buy an article costing ₦400. The woman had 10% of the cost and the man 40% of the remainder. How much did they have altogether?

A. ₦174 B. ₦164 C. ₦184 D. ₦194

109. A pentagon has four of its angles equal. If the size of the fifth angle is 60° , find the size of each of the four equal angles.

A. 120 B. 100 C. 110 D. 130

110. If it is given that $5^{x+1} + 5^x = 150$, then the value of x is equal to

A. 0 B. 1 C. 1.5 D. 2

111. Simplify the given expression $\sqrt{\frac{1-\cos x}{1+\cos x}}$

A. $\frac{1-\cos x}{\sin x}$ B. $1-\cos x$ C. $1+\sin x$ D. $1+\cos x$

112. Write the decimal number 39 to base 2.

A. 110111 B. 100111 C. 111000 <C> 110111

113. Find the smallest number by which 252 can be multiplied to obtain a perfect square

A. 2 B. 3 C. 7 D. 5

114. Find the reciprocal of $\frac{\frac{3}{5}}{\frac{1}{2} + \frac{1}{3}}$

A. $\frac{4}{25}$ B. $\frac{5}{4}$ C. $\frac{2}{3}$ D. $\frac{6}{7}$

115. Divide the L.C.M of 48, 64 and 80 by their H.C.F.

A. 60 B. 30 C. 48 D. 20



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116. The ages of Sola and Akin differ by 6 and the product of their ages is 187. Write their ages in the form (x, y) , where $x > y$.
- A. $(11, 17)$ B. $(11, 16)$ C. $(23, 17)$ D. $(17, 11)$
117. If $5^{(x+2y)} = 5$ and $4^{(x+3y)} = 16$, find $3^{(x+y)}$
- A. 1 B. 0 C. 2 D. 3
118. Find the values of x which satisfy the equation $16^x - 5 + 4^x + 4 = 0$
- A. 0 and -1 B. 1 and 2 C. 0 and 2 D. 0 and 1
119. Factorise $x^2 + 2a + ax + 2x$
- A. $(x+2a)(x+1)$ B. $(x-2a)(x+1)$ C. $(x+2a)(x-1)$
D. $(x+2)(x+a)$
120. An open rectangular box externally measures 4m x 3m x 4m. Find the cost of painting the box externally if its cost ₦2.00 to paint one square metre
- A. ~~₦~~116.00 B. ~~₦~~113.00 C. ~~₦~~112.00 <C> ₦136.00
121. Find the probability that a number selected at random from 40 to 50 is a prime
- A. $\frac{3}{10}$ B. $\frac{3}{11}$ C. $\frac{3}{13}$ D. $\frac{4}{11}$
122. If x varies directly as y^3 and $x=2$ when $y=1$, find x when $y=5$.
- A. 200 B. 350 C. 450 D. 250
123. If Musa scored 75 in Biology instead of 57, his average mark in four subjects would have been 60. What was his total mark?
- A. 220 B. 222 C. 322 D. 122
124. A man kept 6 black, 5 brown and 7 purple shirts in a drawer. What is the probability of his picking a purple shirt with his eyes closed?
- A. $\frac{7}{17}$ B. $\frac{7}{19}$ C. $\frac{7}{20}$ D. $\frac{7}{18}$
125. Evaluate $212_3 - 121_3 + 222_3$
- A. 1121_3 B. 1023_3 C. 1020_3 D. 2020_3
126. Simplify $\frac{0.0324 + 0.00064}{0.48 + 0.012}$
- A. 0.0036 B. 0.036 C. 0.36 D. 3.6
127. Find n if $\log 2^4 + \log 2^7 - \log 2^n = 1$



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- A. 24 B. 13 C. 42 D. 14
128. At what points does the straight line $y=2x+1$ intersect the curve $y=2x^2+5x+1$?
- A. $(-2,-3)$ and $(0.5,1)$ B. $(-2,-3)$ and $(0.5,2)$
C. $(2,3)$ and $(0.5,2)$ D. $(1,2)$ and $(3,4)$
129. If $\cos \theta = \frac{a}{b}$, find $1 + \tan^2 \theta$
- A. $\frac{a^2}{b^2}$ B. $\frac{b^2}{a^2}$ C. $1 + a^2$ D. $1 + b^2$
130. If $P=18, Q=21, R=-6$, and $S=-4$, calculate $\frac{(P-Q)^3}{R^3} + S^2$
- A. $\frac{11}{216}$ B. $\frac{11}{316}$ C. $\frac{11}{416}$ D. $\frac{11}{116}$
131. Sola deposited #150.00 in the bank. At the end of 5 years, the simple interest on the principal was #55.00. At what rate per annum was the interest paid?
- A. 6.33% B. 8.33% C. 7.32% D. 7.33%
132. Find the gradient of the line passing through the points $(-2,0)$ and $(0,-4)$
- A. 2 B. -2 C. 3 D. 4
133. At what value of x is the function $y=x^2-2x-3$ minimum?
- A. 2 B. -2 C. -1 D. 1
134. Solve the equation $(x+2)(x+3)=12$
- A. 1,6 B. 3,6 C. -1,6 D. 1,-6
135. Find the two values of y which satisfy the simultaneous equations $3x+y=8, x^2+xy=6$
- A. 1 and 5 B. 2 and 5 C. 0 and 5 D. -1 and 5
136. Find the sum of the 20 terms in an arithmetic progression whose first term is 7 and the last term is 117
- A. 239 B. 1240 C. 1340 D. 1440
137. The angles of a quadrilateral are $5x+30, 4x+60, 60-x$ and $3x+61$. Find the smallest of these angles.
- A. $60-x$ B. $4x+60$ C. $5x+30$ D. $3x+61$
138. If $g(x)=x^2+3x+4$, find $g(x+1) \cdot g(x)$.
- A. $2(x+1)$ B. $2(x-2)$ C. $x+2$ D. $2(x+2)$



139. Find the positive number n , such that thrice its square is equal to twelve times the number.
- A. 1 B. 4 C. -4 D. -1
140. The area of a square is 144 sq cm . Find the length of its diagonal.
- A. $12\sqrt{2} \text{ cm}$ B. 12 cm C. 13 cm D. 14 cm
141. Simplify $\frac{\sqrt{12}-\sqrt{3}}{\sqrt{12}+\sqrt{3}}$
- A. 3 B. 0 C. 16 D. $\frac{1}{3}$
142. If $S = \{x : x^2 = 9, x > 4\}$, then S is equal to
- A. 0 B. $\{0\}$ C. $\emptyset$ D. $\{\emptyset\}$
143. Express the product of 0.0014 and 0.011 in standard form.
- A. 1.54×10^{-5} B. 1.54×10^{-4} C. 1.54×10^{-3} D. 1.54×10^{-2}
144. What value of g will make the expression $4x^2 - 18xy + g$ a perfect?
- A. $\frac{81y^2}{4}$ B. $\frac{9y^2}{4}$ C. $\frac{81y^2}{4}$ D. $\frac{81y^4}{4}$
145. If $x * y = x + y + xy$, find x when $(x * 2) + (x * 3) = 68$
- A. -21 B. 21 C. 12 D. -12
146. Determine $x + y$ if $\begin{pmatrix} 2 & -3 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 8 \end{pmatrix}$
- A. 3 B. 4 C. 7 D. 12
147. Find the minimum value of $x^2 - 3x + 2$ for all real values of x
- A. -0.75 B. 0.75 C. -0.25 D. 1.25
148. If the function $f(x) = x^3 + 2x^2 + qx - 6$ is divisible by $x + 1$, find q .
- A. -5 B. 5 C. -2 D. 2
149. Find the gradient of the curve $y = 2x(x + 3)$ at $x = 1$
- A. 2 B. -2 C. 1 D. -1
150. Integrate $\frac{1}{x} \cos x$ with respect to x
- A. $\ln x + \sin x + k$ B. $\ln x \cdot \sin x + k$ C. $\ln x \cdot \cos x + k$ D. $\ln x \cdot \cos x \cdot k$



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151. Find the value of K if $\frac{K}{\sqrt{3}} = \sqrt{3}$
- A. 3 B. -3 C. 9 D. -9
152. If $\frac{6C_r}{6P_r} = \frac{1}{6}$, find the value of r.
- A. 1 B. 3 C. 3.5 D. 2
153. How many two-digits numbers can be formed from the digits 0,1,2,3 if a digit can be repeated and no number may begin with 0?
- A. 4 B. 6 C. 13 D. 12
154. The lengths of the sides of a right-angled triangle are xcm, $(3x+1)$ cm and $(3x+1)$ cm. Find x.
- A. 12 B. 11 C. 10 D. 9
155. If $y = x \sin x$, find $\frac{dy}{dx}$ when $x = \frac{\pi}{2}$.
- A. -1 B. 0 C. 1 D. 2
156. P(-6,1) and Q(6,6) are the two ends of the diameter of a given circle. Calculate the radius.
- A. 6 units B. 7.5 units C. 6.5 units D. 7 units
157. Find the rate of change of the volume of v of a sphere with respect to its radius r when $r=1$.
- A. 7π B. 9π C. 10π D. 8π
158. If $6P_r = 6$, find the value of $6P_{r+3}$
- A. 33 b. 30 C. 32 D. 31
159. Teams A and B are involved in a game of football. What is the probability that the game ends in a draw?
- A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{1}{4}$ D. $\frac{2}{3}$
160. The range of the data $k+2, k+3, k+4, k+2, k+5, k+3, k+1$ and $k+6$ is
- A. 10 B. 11 C. 12 D. 13
161. If $A = \begin{pmatrix} 2 & 1 \\ -3 & 0 \end{pmatrix}$ and I is a 2×2 unit matrix, evaluate $A^2 - 2A + 4I$
- A. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ B. $\begin{pmatrix} 1 & 0 \\ 3 & 4 \end{pmatrix}$ C. $\begin{pmatrix} 1 & 5 \\ 3 & 4 \end{pmatrix}$ D. $\begin{pmatrix} 1 & 7 \\ 3 & 4 \end{pmatrix}$



162. If the 9th term of an A.P. is five times the 5th term, find the relationship between a and d .
A. $a+2d=0$ B. $a-d=0$ C. $a+2d-1=0$ D. $a+3d=0$
163. Find the maximum value of y in the equation $y=1-2x+3x^2$
A. $\frac{4}{3}$ B. $\frac{4}{5}$ C. $\frac{3}{5}$ D. $\frac{3}{7}$
164. The binary operation $*$ is defined on the set of integers p and q by $p*q=pq+p+q$. Find $2*(3*4)$.
A. 69 B. 49 C. 59 D. 79
165. Given that $Q=\begin{pmatrix} 6 & 0 \\ 4 & 5 \end{pmatrix}$ and $Q+P=\begin{pmatrix} 7 & -2 \\ 6 & 8 \end{pmatrix}$ evaluate determinant of $Q+2P$
A. 120 B. 123 C. 100 D. 90
166. Find the tangent of the acute angle between the lines $2x+y=3$ and $3x-2y=5$
A. 1.25 B. 1.33 C. 2.75 D. -1.75
167. If the maximum value of $y=1+hx+3x^2$ is 13, find h
A. 12 B. 13 C. 14 D. 11
168. If the standard deviation of the set of numbers 3,6,x,7,5 is $\sqrt{2}$, find the least possible value of x .
A. 2 B. 3 C. 5 D. 6
169. Evaluate $\int_{-2}^1 (x-1)^2 dx$
A. 11 B. 9 C. 10 D. 12
170. Find the area bounded by the curve $y=x(2-x)$, the x -axis, $x=0$ and $x=2$.
A. 1.25sq.units B. 1.33sq.units C. 0.33sq.units
D. 2.33sq.units
171. A trader realizes $10x-x^2$ naira profit from the sale of x bags of corn. How many bags will give him the maximum profit?
A. 6 B. 4 C. 3 D. 5
172. If a and b are the roots of the equation $3x^2+5x+2=0$, find the value of $\frac{1}{a} + \frac{1}{b}$.
A. -2.5 B. 0.4 C. 1.5 D. 2.5
173. If $P344_6 - 23P2_6 = 2PP2_6$ find the value of digit P .
A. 4 B. 5 C. 6 D. 7



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174. Find the minimum value of the function $f(\theta) = \frac{2}{3 - \cos \theta}$ for $0 \leq \theta \leq 2\pi$
- A. 0.50 B. 1.67 C. 0.67 D. 2.67
175. X and Y are two events. The probability of X or Y is 0.7 and the probability of X is 0.4. If X and Y are independent, find the probability of Y.
- A. 0.2 B. 0.4 C. 0.5 D. 0.3
176. An equilateral triangle of side 3cm is inscribed in a circle. Find the radius of the circle.
- A. 1.0cm B. 2.0cm C. 3.0cm D. 0.7cm
177. In a class of 40 students, 32 offer Mathematics, 24 offer Physics and 4 offer neither Mathematics nor Physics. How many offer both Mathematics and Physics?
- A. 16 B. 21 C. 19 D. 20
178. If $\frac{9^{2x-1}}{27^{3x-8}} = 1$, find the value of x.
- A. 3 B. 5 C. 6 D. 7
179. If $\begin{vmatrix} -x & 2 \\ 4x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 3x \\ 4 & -5 \end{vmatrix}$, find the value of x
- A. 5 B. 4 C. 3 D. .5
180. A cinema hall contains a certain number of people. If 22.5% are children, 47.5% are men and 84 are women, find the number of men in the hall.
- A. 133 B. 132 C. 130 D. 123
181. Find the value of p, if the line which passes through $(-1, p)$ and $(-2p, 2)$ is parallel to the line $2y + 8x - 17 = 0$.
- A. $\frac{6}{5}$ B. $\frac{6}{7}$ C. $\frac{6}{11}$ D. $\frac{7}{11}$
182. An arc of a circle subtends an angle of 30° on the circumference of a circle of radius 21cm. Find the length of the arc.
- A. 11cm B. 22cm C. 66cm D. 44cm
183. Find the remainder when $3x^3 + 5x^2 - 11x + 4$ is divided by $x + 3$.
- A. -1 B. 2 C. 4 D. 1
184. The n^{th} terms of two sequences are $Q_n = 3x2^{n-2}$ and $U_m = 3x2^{2m-3}$. Find the product of Q_2 and U_2
- A. 18 B. 16 C. 6 D. 3



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185. If the operation $*$ on the set of integers is defined by $p * q = \sqrt{pq}$, find the value of $4 * (8 * 32)$.
- A. 16 B. 8 C. 6 D. 18
186. Find the sum to infinity of the series $\frac{1}{2}, \frac{1}{6}, \frac{1}{12}, \dots$
- A. 1 B. 0.25 C. 0.75 D. 1.75
187. A man 40m from the foot of a tower observes the angle of elevation of the tower to be 30° . Determine the height of the tower.
- A. $\frac{40\sqrt{3}}{3}$ m B. 40m C. 20m D. $40\sqrt{3}$ m
188. A cliff on the bank of a river is 300m high. If the angle of depression of a point on the opposite side of the river is 60° , find the width of the river.
- A. 100m B. 150m C. $100\sqrt{3}$ cm D. 200m
189. The mean of a set of six numbers is 60. If the mean of the first five is 50, find the sixth number in the set.
- A. 100 B. 120 C. 105 D. 110
190. Make r the subject of the formula $\frac{x}{a+r} = \frac{a}{r}$
- A. $\frac{a}{a+r}$ B. $\frac{a^2}{x-a}$ C. $\frac{a^2}{x+a}$ <C> D. $\frac{a}{a-r}$
191. The inverse of the function $f(x) = 3x + 4$ is
- A. $\frac{x-4}{3}$ B. $\frac{x+4}{3}$ C. $\frac{3}{x-4}$ D. $\frac{3}{x+4}$
192. If $\frac{dy}{dx} = 2x - 3$ and $y = 3$ when $x = 0$, find y in terms of x
- A. $x^2 - 3x - 3$ B. $x^2 - 3x + 3$ C. $x^2 + 3x - 3$
- D. $x^2 + 3x + 3$
193. A circle with a radius 5cm has its radius increasing at the rate of 0.2cm/s. What will be the corresponding increase in the area?
- A. 3π B. 4π C. 2π D. 5π
194. Find the range of values of x for which $\frac{x+2}{4} - \frac{2x-3}{3} < 4$
- A. $x < 6$ B. $x > 6$ C. $x < -6$ D. $x > -6$
195. If -2 is the solution of the equation $2x + 1.3c = 2c + 3x + 7$, find the value of c
- A. 2 B. -2 C. 3 D. -3



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196. The sum of the interior angles of a regular polygon is 1800° . Calculate the size of one exterior angle of the polygon.
- A. 45° B. 60° C. 30° D. 90°
197. Find the simple interest rate percent per annum at which #1,000 accumulates to #1,240 in 3 years.
- A. 8% B. 7% C. 6% D. 5%
198. Three consecutive positive integers k , l and m are such that $l^2 = 3(k+m)$. Find the value of m .
- A. 4 B. 5 C. 6 D. 7
199. Find the value of x if $\frac{\sqrt{x}}{x+\sqrt{x}} = \frac{1}{x-\sqrt{x}}$
- A. $3\sqrt{2}-4$ B. $3\sqrt{2}+4$ C. $3\sqrt{2}-3$ D. $3\sqrt{2}+3$
200. The expression $ax^2 + bx + c$ equals 5 at $x=1$. If its derivative is $2x+1$, what are the values of a, b, c respectively.
- A. 1, 3, 1 B. 1, -3, 1 C. 1, 1, 3 D. 1, 3, -1
201. If $\tan \theta = \frac{5}{4}$, find $\sin^2 \theta - \cos^2 \theta$
- A. $\frac{41}{9}$ B. $\frac{41}{3}$ C. $\frac{9}{41}$ D. $\frac{19}{41}$
202. If $2q3s = 77s$, find q .
- A. -2 B. 3 C. 2 D. 4
203. Simplify $\frac{\frac{3}{5} \times \frac{5}{6} \times \frac{6}{7}}{\frac{7}{8} \times \frac{8}{9} \times \frac{9}{10}}$
- A. 50 B. 30 C. 45 D. 35
204. A man invested #5000 for 9 months at 4%. What is the simple interest?
- A. ₦ 220 B. ₦ 130 C. ₦ 150 D. ₦ 250
205. If the numbers M, N, Q are in the ratio 5:4:3, find the value of $\frac{2N-Q}{M}$.
- A. 1 B. 2 C. 4 D. 31
206. Simplify $\left(\frac{16}{81}\right)^{\frac{1}{4}} + \left(\frac{9}{16}\right)^{\frac{-1}{2}}$
- A. $\frac{2}{3}$ B. $\frac{1}{2}$ C. $\frac{8}{9}$ D. $\frac{1}{3}$



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207. If $\log_3 18 + \log_3 3 + \log_3 x = 3$, find x .
A. 2 B. 1 C. 0 D. 3
208. Rationalize $\frac{2-\sqrt{5}}{3-\sqrt{5}}$
A. $\frac{1-\sqrt{5}}{2}$ B. $\frac{1-\sqrt{5}}{3}$ C. $\frac{1+\sqrt{5}}{4}$ D. $\frac{1-\sqrt{5}}{4}$
209. Simplify $\left(\sqrt{2} + \frac{1}{\sqrt{3}}\right)\left(\sqrt{2} - \frac{1}{\sqrt{3}}\right)$
A. $\frac{7}{3}$ B. $\frac{5}{3}$ C. $\frac{5}{2}$ D. $\frac{3}{2}$
210. Raila has 7 different posters to be hanged in her bedroom, living room and kitchen. Assuming she has plans to replace at least a poster in each of the 3 rooms, how many choices does she have?
A. 49 B. 170 C. 210 D. 21
211. Find the remainder when $x^3 + 2x^2 + 3x + 3$ is divided by $x^2 + 1$.
A. $x+3$ B. $2x+1$ C. $2x+1$ D. $x+3$
212. Factorize completely $9y^2 + 16x^2$.
A. $(3y+2x)(3y+4x)$ B. $(3y+4x)(3y+4x)$
C. $(3y+2x)(3y+4x)$ D. $(3y+4x)(3y+4x)$
213. Solve for x and y respectively in the simultaneous equations $-2x + 5y = 3$, $x + 3y = 0$.
A. $-9, 3$ B. $9, -3$ C. $3, -9$ D. $-3, -9$
214. If x varies directly as square root of y and $x = 81$ when $y = 9$, find x when $y = 1\frac{7}{9}$.
A. 27 B. 20.25 C. 36 D. 2.25
215. T varies inversely as the cube of R . When $R = 3$, $T = \frac{2}{81}$, find T when $R = 2$.
A. $\frac{1}{18}$ B. $\frac{1}{12}$ C. $\frac{1}{24}$ D. $\frac{1}{6}$
216. Solve the inequality $-6(x+3) \leq 4(x-2)$.
A. $x \leq 2$ B. $x \leq -2$ C. $x \leq -1$ D. $x \geq -1$
217. Solve the inequality $x^2 + 2x > 15$.
A. $x > 3$ or $x < -5$ B. $x < -3$ or $x > 5$ C. $-5 < x < 3$ D. $x < 3$ or $x > 5$
218. Find the sum of the first 18 terms of the series 3, 6, 9, ..., 36



A. 513 B. 505 C. 433 D. 635

219. The second term of a geometric series is 4 while the fourth term is 16. Find the sum of the first five terms.

A. 60 B. 54 C. 64 D. 62

220. A binary operation $*$ on real numbers is defined by $x * y = xy + x + y$ for two real numbers x and y . Find the value of $3 * -\frac{2}{3}$.

A. $\frac{2}{3}$ B. $\frac{1}{3}$ C. -1 D. 2

221. If $\begin{vmatrix} 2 & 3 \\ 5 & 3x \end{vmatrix} = \begin{vmatrix} 4 & 1 \\ 3 & 2x \end{vmatrix}$, find the value of x .

A. -6 B. 6 C. 12 D. -12

222. Evaluate $\begin{vmatrix} 4 & 2 & -1 \\ 2 & 3 & -1 \\ -1 & 1 & 3 \end{vmatrix}$

A. 45 B. 15 C. 55 D. 25

223. The inverse of matrix $N = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ is

A. $\frac{1}{5} \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$ B. $\frac{1}{5} \begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix}$ C. $\frac{1}{5} \begin{pmatrix} 2 & -1 \\ -3 & 4 \end{pmatrix}$
D. $\frac{1}{5} \begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix}$

224. What is the size of each interior angle of a 12-sided regular polygon?

A. 120° B. 150° C. 30° D. 180°

225. A circle of perimeter 28cm is opened to form a square. What is the maximum possible area of the square?

A. 56cm^2 B. 98cm^2 C. 49cm^2 D. 28cm^2

226. A chord of a circle of radius 7cm is 5cm from the centre of the circle. What is the length of the chord?

A. $4\sqrt{6}\text{cm}$ B. $3\sqrt{6}\text{cm}$ C. $6\sqrt{6}\text{cm}$ D. $2\sqrt{6}\text{cm}$

227. A solid metal cube of side 3cm is placed in a rectangular tank of dimensions 3, 4 and 5 cm. What volume of water can the tank now hold?

A. 48cm^3 B. 33cm^3 C. 60cm^3 D. 27cm^3

228. The perpendicular bisector of a line XY is the locus of a point

A. whose distance from X is always twice its distance from Y



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- B whose distance from Y is always twice its distance from X
- C which moves on the line XY
- D which is equidistant from the points X and Y
- 229 The midpoint of P(x, y) and Q(8, 6) is (5, 8). Find x and y.
- A (2, 10) B. (2, 8) C. (2, 12) D. (2, 6)
- 230 Find the equation of a line perpendicular to line $2y - 5x + 4$ which passes through (4, 2).
- A. $5y - 2x - 18 = 0$ B. $5y + 2x - 18 = 0$ C. $5y - 2x + 18 = 0$
- D. $5y + 2x + 2 = 0$
- 231 In a right angled triangle, if $\tan \theta = \frac{3}{4}$. What is $\cos \theta - \sin \theta$?
- A. $\frac{1}{4}$ B. $\frac{3}{5}$ C. $\frac{1}{5}$ D. $\frac{2}{5}$
- 232 A man walks 100m due West from a point X to Y, he then walks 100m due North to a point Z. Find the bearing of X from Z.
- A. 195° B. 135° C. 225° D. 045°
- 233 The derivative of $(2x+1)(3x+1)$ is
- A. $12x+1$ B. $6x+5$ C. $6x+1$ D. $12x+5$
- 234 Find the value of x at the minimum point of the curve $y = x^3 + x^2 \cdot x + 1$.
- A. $\frac{1}{3}$ B. $-\frac{1}{3}$ C. 1 D. -1
- 235 Evaluate $\int_0^1 (3 - 2x) dx$.
- A. 2 B. 5 C. 6 D. 3
- 236 Find $\int \cos 4x dx$.
- A. $\frac{3}{4} \sin 4x + k$ B. $-\frac{1}{4} \sin 4x + k$ C. $\frac{1}{4} \sin 4x + k$
- D. $-\frac{3}{4} \sin 4x + k$
- 237 The sum of four consecutive integers is 34. Find the least of these numbers.
- A. 6 B. 8 C. 7 D. 5



No.	0	1	2	3	4	5
Frequency	1	4	3	8	2	5

238 find the median and range of the data respectively.

- A. (8, 5) B. (3, 5) C. (5, 8) D. (5, 3)

Class Interval	0-2	3-5	6-8	9-11
Frequency	1	4	3	8

239 Find the mode of the above distribution.

- A. 9 B. 8 C. 10 D. 7

Class Interval	3-5	6-8	9-11
Frequency	2	2	2

240 Find the standard deviation of the above distribution

- A. $\sqrt{3}$ B. $\sqrt{5}$ C. $\sqrt{7}$ D. $\sqrt{2}$

241 In how many ways can the letters of the word ELATION be arranged?

- A. 6! B. 5! C. 8! D. 7!

242 In how many ways can five people sit round a circular table?

- A. 60 B. 24 C. 12 D. 120

243 Find the probability that a number picked at random from the set {43, 44, 45, ..., 60} is a prime number.

- A. $\frac{2}{3}$ B. $\frac{2}{9}$ C. $\frac{1}{3}$ D. $\frac{7}{9}$

244 In a class of 60 students, 30 offer Physics and 40 offer Chemistry. If a student is picked at random from the class, what is the probability that the student offer both Physics and Chemistry?

- A. $\frac{1}{3}$ B. $\frac{1}{4}$ C. $\frac{1}{2}$ D. $\frac{1}{6}$



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- 245 Convert 72_6 to a number in base three.
A. 2211 B. 2121 C. 1212 D. 1122
- 246 Simply $\frac{2\frac{2}{3} \times 1\frac{1}{2}}{4\frac{4}{5}}$
A. $1\frac{2}{4}$ B. $1\frac{1}{6}$ C. $\frac{5}{6}$ D. $\frac{4}{3}$
- 247 Evaluate $\frac{21}{9}$ to 3 significant figures.
A. 2.30 B. 2.31 C. 2.32 D. 2.33
- 248 A man earns ? 3 500 per month out of which he spends 15% on his children's education. If he spends additional ? 1 950 on food, how much does he have left?
A. ? 525 B. ? 1 025 C. ? 1 950 D. ? 2 975
- 249 If $27^{x+2} + 9^{x+1} = 3^{2x}$ find x .
A. 3 B. 4 C. 5 D. 6
- 250 If $\log_3 x^2 = -8$, what is x ?
A. $\frac{1}{3}$ B. $\frac{1}{9}$ C. $\frac{1}{27}$ D. $\frac{1}{81}$
- 251 Simplify $(\sqrt{6} + 2)^2 - (\sqrt{6} - 2)^2$.
A. $2\sqrt{6}$ B. $4\sqrt{6}$ C. $8\sqrt{6}$ D. $16\sqrt{6}$
- 252 If P is a set of all prime factors of 30 and Q is a set of all factors of 18 less than 10, find $P \cap Q$.
A. {3} B. {2,3} C. {2,3,5} D. {1,2}
- 253 In a class of 46 students, 22 play football and 26 play volleyball. If 3 students play both games, how many play neither?
A. 1 B. 2 C. 3 D. 4
- 254 Make n the subject of the formula if $w = \frac{v(2 + cn)}{1 - cn}$
A. $\frac{1}{c} \left(\frac{w - 2v}{v + w} \right)$ B. $\frac{1}{c} \left(\frac{w - 2v}{v - w} \right)$ C. $\frac{1}{c} \left(\frac{w + 2v}{v - w} \right)$
D. $\frac{1}{c} \left(\frac{w + 2v}{v + w} \right)$



255 Find the remainder when $2x^3 - 11x^2 + 18x - 1$ is divided by $x + 3$.

- A. -871 B. -781 C. -187 D. -178

256 Solve for x and y in the equation below.

$$x^2 - y^2 = 4$$

$$x + y = 2$$

- A. $x = 0, y = -2$ B. $x = 0, y = 2$ C. $x = 2, y = 0$
D. $x = -2, y = 0$

257 If y varies directly as $\sqrt{n}$ and $y = 4$ when $n = 4$, find y when $n = 1\frac{7}{9}$.

- A. $\sqrt{17}$ B. $\frac{4}{3}$ C. $\frac{8}{3}$ D. $\frac{2}{3}$

258 U is inversely proportional to the cube of V and $U = 81$ when $V = 2$. Find U when $V = 3$.

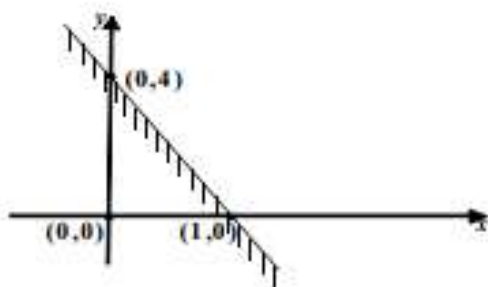
- A. 24 B. 27 C. 32 D. 36

259 The value of y for which $\frac{1}{5}y + \frac{1}{5} < \frac{1}{2}y + \frac{2}{5}$ is

- A. $y > \frac{2}{3}$ B. $y < \frac{2}{3}$ C. $y > -\frac{2}{3}$ D. $y < -\frac{2}{3}$

260 Find the range of values of m which satisfies $(m - 3)(m - 4) < 0$.

- A. $2 < m < 5$ B. $-3 < m < 4$ C. $3 < m < 4$
D. $-4 < m < 3$



261 The shaded region above is represented by the equation.

- A. $y \leq 4x + 2$ B. $y \geq 4x + 2$ C. $y \leq -4x + 4$
D. $y \leq 4x + 4$



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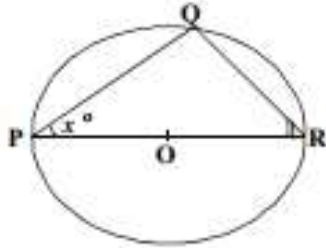
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- 262 The n th term of a sequence is $n^2 - 6n - 4$. Find the sum of the 3rd and 4th terms.
A. 24 B. 23 C. -24 D. -25
- 263 The sum to infinity of a geometric progression is $-\frac{1}{10}$ and the first term is $-\frac{1}{8}$. Find the common ratio of the progression.
A. $-\frac{1}{5}$ B. $-\frac{1}{4}$ C. $-\frac{1}{3}$ D. $-\frac{1}{2}$
- 264 The binary operation $*$ is defined on the set of integers such that $p * q = pq + p - q$. Find $2 * (3 * 4)$.
A. 11 B. 13 C. 15 D. 22
- 265 A binary operation on the set of real numbers is defined by $m * n = \frac{mn}{2}$ for all $m, n \in R$. If the identity element is 2, find the inverse of -5.
A. $-\frac{4}{5}$ B. $-\frac{2}{5}$ C. 4 D. 5
- 266 If $\begin{vmatrix} 5 & 3 \\ x & 2 \end{vmatrix} = \begin{vmatrix} 3 & 5 \\ 4 & 5 \end{vmatrix}$, find the value of x
A. 3 B. 4 C. 5 D. 7
- 267 Given that I_3 is a unit matrix of order 3, find $|I_3|$
A. -1 B. 0 C. 1 D. 2
- 268
-
- In the diagram above, $QR \parallel TU$, $\angle PQR = 80^\circ$ and $\angle PSU = 95^\circ$. Calculate $\angle SUT$.
A. 15° B. 25° C. 30° D. 80°
- 269 The angles of a polygon are given by x , $2x$, $3x$, $4x$ and $5x$ respectively. Find the value of x
A. 24° B. 30° C. 33° D. 36°

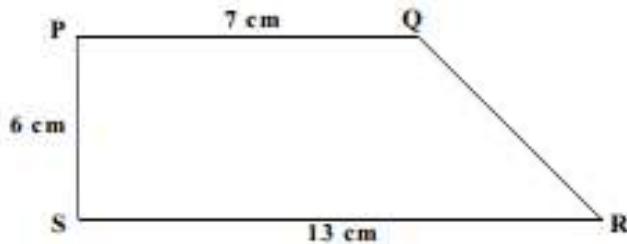
270



In the diagram above, PQR is a circle centre O . If $\angle QPR$ is x° , find $\angle QRP$.

- A. x° B. $(90 - x)^\circ$ C. $(90 + x)^\circ$ D. $(180 - x)^\circ$

271



Find the area of the trapezium above.

- A. 91 cm^2 B. 78 cm^2 C. 60 cm^2 D. 19 cm^2

272

A circular arc subtends angle 150° at the centre of a circle of radius 12 cm . Calculate the area of the sector of the arc.

- A. $30\pi \text{ cm}^2$ B. $60\pi \text{ cm}^2$ C. $120\pi \text{ cm}^2$ D. $150\pi \text{ cm}^2$

273

Calculate the volume of a cuboid of length 0.76 cm , breadth 2.6 cm and height 0.82 cm .

- A. 3.92 cm^3 B. 2.13 cm^3 C. 1.97 cm^3 D. 1.62 cm^3

274

The locus of a point equidistant from the intersection of lines $3x - 7y + 7 = 0$ and $4x - 6y + 1 = 0$ is a

- A. line parallel to $7x - 13y + 8 = 0$ B. circle
C. semicircle D. bisector of the line $7x - 13y + 8 = 0$.

275

The gradient of the straight line joining the points $P(5, -7)$ and $Q(-2, -3)$ is

- A. $\frac{1}{2}$ B. $\frac{2}{3}$ C. $-\frac{4}{7}$ D. $-\frac{2}{3}$

276

The distance between the point $(4, 3)$ and the intersection of $y = 2x + 4$ and $y = 7 - x$ is



- A. $\sqrt{13}$ B. $3\sqrt{2}$ C. $\sqrt{26}$ D. $10\sqrt{5}$

277 Find the equation of the lines through the points $(-2, 1)$ and $(-\frac{1}{2}, 4)$

- A. $y = 2x - 3$ B. $y = 2x + 5$ C. $y = 3x - 2$
D. $y = 2x + 1$

278 If angle θ is 135° , evaluate $\cos \theta$.

- A. $\frac{1}{2}$ B. $\frac{\sqrt{2}}{2}$ C. $-\frac{\sqrt{2}}{2}$ D. $-\frac{1}{2}$

279 A man stands on a tree 150 cm high and sees a boat at an angle of depression of 74° . Find the distance of the boat from the base of the tree.

- A. 52 cm B. 43 cm C. 40 cm D. 15 cm

280 If $y = x^2 - \frac{1}{x}$, find $\frac{dy}{dx}$.

- A. $y = 2x - \frac{1}{x^2}$ B. $2x + x^2$ C. $2x - x^2$
D. $2x + \frac{1}{x^2}$

281 Find $\frac{dy}{dx}$, if $y = \cos x$.

- A. $\sin x$ B. $-\sin x$ C. $\tan x$ D. $-\tan x$

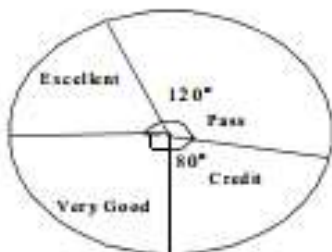
282 Evaluate $\int_1^2 (x^2 - 4x) dx$.

- A. $\frac{11}{3}$ B. $\frac{3}{11}$ C. $-\frac{3}{11}$ D. $-\frac{11}{3}$

283 Evaluate $\int_0^{\frac{\pi}{4}} (\sec^2 \theta) d\theta$.

- A. 1 B. 2 C. 3 D. 4

284

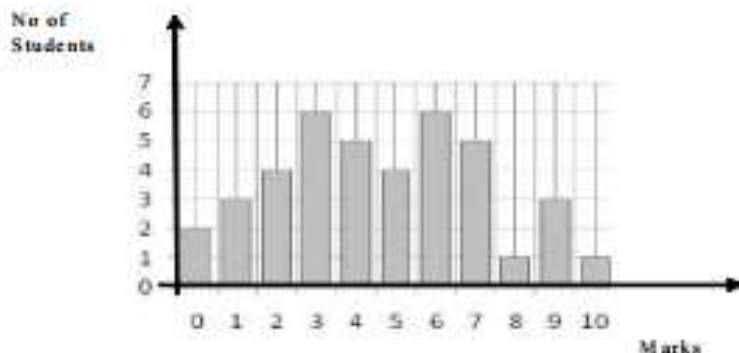




The grades of 36 students in a class test are as shown in the pie chart above. How many students have excellent?

- A. 12 B. 9 C. 8 D. 7

285



The bar chart above shows the distribution of marks in a class test. If the pass mark is 5, what percentage of the students failed the test?

- A. 10% B. 20% C. 50% D. 60%

286 The mean of seven numbers is 96. If the eight number is added, the mean becomes 112. Find the eight number.

- A. 126 B. 180 C. 216 D. 224

287 Find the median of 2,3,7,3,4,5,8,9,9,4,5,3,4,2,4 and 5

- A. 9 B. 8 C. 7 D. 4

288 Find the range of 4,9,6,3,2,8,10 and 11.

- A. 11 B. 9 C. 8 D. 4

289 Find the standard deviation of 2,3,8, 10 and 12.

- A. 3.9 B. 4.9 C. 5.9 D. 6.9

290 Evaluate ${}^{n+1}C_{n-2}$ If $n = 15$.

- A. 3630 B. 3360 C. 1120 D. 560

291 In how many ways can the letters of the word TOTALITY be arranged?

- A. 6720 B. 6270 C. 6207 D. 6027

292 The probability that a student passes a physics test is $\frac{2}{3}$, If he takes three physics test, what is the probability that he passes two of the test.



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- A. $\frac{4}{9}$ B. $\frac{6}{9}$ C. $\frac{4}{27}$ D. $\frac{2}{27}$

293. The probability that a man and his wife live for 80 years are $\frac{2}{3}$ and $\frac{3}{5}$ respectively. Find the probability that at least one of them will live up to 80 years.

- A. $\frac{2}{15}$ B. $\frac{3}{15}$ C. $\frac{7}{15}$ D. $\frac{13}{15}$

294. Solve the equation $\log_8 x + 4 \log_8 x = 2$

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) 1 (d) 2 (e) 4

295. Find the product of x and y if $x = \frac{2}{3}$, $\frac{6}{7}$, y are in GP

- (a) $\frac{24}{49}$ (b) $\frac{4}{7}$ (c) $\frac{2}{7}$ (d) $\frac{7}{4}$ (e) $2\frac{1}{8}$

296. A box contains 2 white and 3 blue identical balls.

If two balls are picked at random one after the other without replacement, what is the probability of picking two balls of different colours?

- (a) $\frac{6}{25}$ (b) $\frac{7}{20}$ (c) $\frac{3}{5}$ (d) $\frac{2}{3}$ (e) $\frac{5}{6}$

297. For what value of y is the expression $y^2 + 2$ undefined? $Y^2 - 3y - 10$

- (a) $Y = 0$ (b) $y = 2$ (c) $y = 3$ (d) $y = 5$ (e) $y = 10$

298. A piece of cloth was measured as 6.10m. If the actual length of the cloth is 6.35m, find the percentage error correct to 2 decimal places.

- (a). 3.05% (b) 3.94% (c) 15.00% (d) 25.00% (E) 39.30%

299. The radius of a geographical globe is 60cm. Find the length of the parallel of latitude $60^\circ N$

- (a) 66cm (b) 60cm (c) 30cm (d) 10cm (e) 6cm

300. A die with faces numbered 1 to 6 is rolled once. What is the probability of obtaining 1 or 4?

- (a) $\frac{1}{36}$ (b) $\frac{1}{6}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$ (e) $\frac{2}{3}$



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1.B 2.C 3.C 4.A 5.B 6.A 7.C 8.B 9.D 10.A 11.A 12.D 13.D
4.B 15.A 16.D 17.C 18.B 19.D 20.B 21.A 22.C 23.D 24.D 25.A 26.A
27.A 28.B 29.A 30.C 31.A 32.D 33.D 34.C 35.A 36.B 37.B 38.D 39.B
40.C 41. 42.A 43.B 44.B 45.B 46.B 47.B 48.C 49.B 50.C 51.C 52.B
53.C 54.C 55.C 56.B 57.D 58.D 59.C 60.C 61.C 62.A 63.A 64.A 65.B
66.C 67.D 68.C 69.D 70.C 71.A 72.C 73.C 74.A 75.C 76.B 77.A 78.C
79.A 80.A 81.D 82.D 83.A 84.B 85. 86.C 87.B 88.A 89.D 90.A 91.C
92.D 93.B 94.A 95.A 96.B 97.A 98.A 99.A 100.A 101.D 102.D 103.C 104.D
105.B 106.A 107.D 108.C 109.A 110.D 111.A 112.B 113.C 114.B 115.A 116.D 117.A
118.D 119.D 120.C 121.B 122.D 123.B 124.C 125.C 126.A 127.D 128.B 129.B 130.A
131.D 132.B 133.D 134.C 135.D 136.B 137.A 138.D 139.B 140.A 141.D 142.C 143.A
144.C 145.A 146.C 147.C 148.A 149.B 150.A 151.A 152.B 153.D 154.A 155.C 156.C
157.D 158.B 159.C 160.B 161.A 162.D 163.A 164.C 165.A 166.D 167.A 168.D 169.B
170.B 171.D 172.D 173.B 174.C 175.D 176.A 177.D 178.B 179.D 180.A 181.B 182.B
183.D 184.A 185.B 186.C 187.A 188.C 189.D 190.B 191.A 192.B 193.C 194.D 195.A
196.C 197.A 198.D 199.B 200.C 201.A 202.C 203.A 204.C 205.A 206.B 207.A 208.D
209.B 210.D 211.B 212.D 213.A 214.C 215.B 216.D 217.A 218.A 219.D 220.B 221.A
222.D 223.B 224.B 225.C 226.A 227.B 228.D 229.C 230.B 231.C 232.B 233.D 234.A
235.A 236.C 237.C 238.B 239.D 240.A 241.D 242.B 243.B 244.D 245.D 246.C 247.D
248.A 249.B 250.D 251.C 252.B 253.A 254.A 255.D 256.C 257.C 258.A 259.C 260.C
261.C 262.D 263.B 264.B 265.A 266.C 267.B 268.A 269.D 270.B 271.C 272.B 273.D
274.B 275.C 276.B 277.B 278.C 279.B 280.D 281.B 282.D 283.A 284.D 285.C 286.D
287.D 288.B 289.A 290.D 291.A 292.A 293. 294.A 295.A 296.C 297.D 298.B 299.E 300.C



CONTINUATION

1. What is the probability of having an odd number in a single toss of a fair die?
(a) $1/6$ (b) $1/3$ (c) $1/2$ (d) $2/3$ (e) $5/6$
2. A sales girl gave a change of N 1.15 to a customer instead of N 1.25. Calculate the Percentage error. (a) 10% (b) 8.7% (c) 8.0% (d) 2.4% (e) 0.10%
- 3.

Score	0	1	2	3	4	5
Frequency	2	3	4	2	7	2

Find the mean score in the frequency table above showing the performance a group of student in a test.

- (a) 1.75 (b) 2 (c) 2.5 (d) 2.75 (e) 3
4. Simplify $0.63954/0.003$ giving your answer correct to two significant figures.
(a) 213.18 (b) 213.00 (c) 213 (d) 210 (e) 2
5. What is the difference in longitude between P (lat. 50°N , long. 50°W) and Q (lat. 50°N , long. 150°W)
(a) 300° (b) 200° (c) 130° (d) 100° (e) 300°
6. Simplify $(27^{1/3})^2$. (a) $4\frac{1}{2}$ (b) 6 (c) 9 (d) 18 (e) 81
7. If $\log_{10} a = 4$, what is a? (a) 0.4 (b) 40 (c) 400 (d) 1000 (e) 10000
8. Simplify $36^{1/2} \times 64^{-1/2} \times 5^0$ (a) 0 (b) $1/24$ (c) $2/3$ (d) $11/2$ (e) 7 $\frac{1}{2}$
9. Solve the equation $3a + 10 = a^2$ (a) $a = 5$ or $a = 2$ (b) $a = -5$ or $a = 2$ (c) $a = 10$ or $a = 0$ (d) $a = 5$ or $a = -2$ (e) $a = -5$
10. Solve $6(x-4) + 3(x+7) = 3$ (a) $3/2$ (b) $2/3$ (c) $1/2$ (d) $1/3$ (e) 0
11. The common ratio of a GP is 2. If the 5th term is greater than the 1st term by 45 find the 5th term.
(a) 3 (b) 6 (c) 45 (d) 48 (e) 90
12. A house bought for N100, 000 was later auctioned for N 80,000. Find the loss percent. (A) 20% (b) 30% (c) 40% (d) 50% (e) 60%
13. Simplify $2 \log_3 6 + \log_3 12 - \log_3 16$ (A) 2 (b) 3 (c) $22 \log_3 2$ (d) $3 \log_3 2$ (e) $4 \log_3 2$
14. If $8x - 4 = 6x - 10$, find the value of $5x$. (a) -35 (b) -15 (c) -3 (d) 3 (e) 7
15. Given that $2p - m = 6$ and $2p + 4m = 1$. Find the value of $(4p + 3m)$. (a) 1 (b) 3 (c) 5 (e) 9
16. Which of the following is not a factor of $2p^2 - 2$. (a) $2(p-1)$ (c) $p+1$ (d) $2p-2$ (e) $2p+1$
17. A car is traveling at an average speed of 80 km/hr. Its speed in meters per Second (m/s) is (a) 22.2m/s (b) 133.3m/s (c) 13.3m/s (d) 222.2m/s



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18. Solve the equation $\log_8 x^4 - \log_8 x = 2$ (a) $1/4$ (b) $1/2$ (c) 1 (d) 2 (e) 4
19. Find the product of x and y if $x = 2/3$, $6/7$, y are in GP (a) $24/49$ (b) $4/7$ (c) $2/7$ (d) $7/4$ (e) $2 \frac{1}{8}$
20. A box contains 2 white and 3 blue identical balls. If two balls are picked at random one after the other without replacement, what is the probability of picking two balls of different colours? (a) $6/25$ (b) $7/20$ (c) $3/5$ (d) $2/3$ (e) $5/6$
21. For what value of y is the expression $y^2 - 3y - 10$ undefined? (a) $Y = 0$ (b) $y = 2$ (c) $y = 3$ (d) $y = 5$ (e) $y = 10$
22. A piece of cloth was measured as 6.10m. If the actual length of the cloth is 6.35m, find the percentage error correct to 2 decimal places. (a) 3.05% (b) 3.94% (c) 15.00% (d) 25.00% (e) 39.30%
23. The radius of a geographical globe is 60cm. Find the length of the parallel of latitude $60^\circ N$ (a) 66cm (b) 60cm (c) 30cm (d) 10cm (e) 6cm
24. A die with faces numbered 1 to 6 is rolled once. What is the probability of obtaining 1 or 4? (a) $1/36$ (b) $1/6$ (c) $1/3$ (d) $1/2$ (e) $2/3$
25. Make S the subject of the formula $V = K/\sqrt{T-S}$ (a) $S = T - (K^2/V^2)$ (b) $S(K^2/V^2) = T$ (c) $S = T - (K^2/V^2)$ (d) $S = T(V^2 - K^2)/K^2$ (e) $S = T + (V^2/K^2)$

ANSWERS

1.C 2.C 3.D 4.E 5.D 6.C 7.E 8.C 9.B 10.B 11.D 12.A 13.B 14.B 15.E 16.E 17.A 18.A 19.A 20.C
21.D 22.B 23.E 24.C 25.C

1. The factors $6x^2 - 5x$ are (a) $(x+3), (x+2)$ (b) $(x-5), (x-1)$ (c) $-(x+5), (x+1)$ (d) $(x-5), (1-x)$ (e) $(x+5), (1-x)$
2. Seven years ago, the age of a father was three times that of his son, but in six years time the age of the son will be half of the father. Representing the ages of the father and son by X and Y respectively, the two equations relating X and Y are: (a) $3Y - X = 0; 2Y - X = 0$ (b) $3Y - X = 14; X - 2Y = 6$ (c) $3Y - X = 7; X - 2Y = 6$ (d) $3Y - X = 14; 2Y - X = 6$ (e) $X + 3Y = 7; X - 2Y = 12$
3. If M represents the median and D is the mode of the measurements 5, 9, 3 and 5, the (M, D) is (a) (6, 5) (b) (5, 8) (c) (5, 7) (d) (5, 5) (e) (7, 5)
4. The value of $(0.03)^2 - (0.02)^2$ is (a) 0.019 (b) 0.0019 (c) 0.00019 (d) 0.000019 (e) 0.000035
5. If $263 + 441 = 714$, what number base has been used? (a) 12 (b) 11 (c) 10 (d) 9 (e) 8



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6. Bala chooses at random a number between 1 and 300. What is the Probability that the number is divisible by 4 of the father and son by X and Y, respectively, the two equations relating X and Y are; a. $1/3$ b. $1/4$ c. $1/5$ d. $4/300$ e. $1/300$
7. A piece of cloth was measured as 6.10m. If the actual length of the cloth is 6.35m, find the percentage error correct to 2 decimal places. (a) 3.05% (b) 3.94% (c) 15.00% (d) 25.00% (E) 39.30%
8. Simplify $2\log_3 6 + \log_3 12 - \log_3 16$ (a) 2 (b) 3 (c) $22\log_3 2$ (d) $3\log_3 2^3$ (e) $4\log_3 2$
9. Find correct to two decimal places $100 + 1/100 + 3/1000 + 27/10000$ a. 100.02 b. 1000.02 c. 100.22 d. 100.01 e. 100.51
10. A bag contains 5 black, 4 white and x red marbles. If the probability of picking a red marble is $2/5$, find the value of x. a. 8 b. 10 c. 4 d.) 6
11. Simplify $0.0324 \times 0.0064 / 0.48 \times 0.012$ (a) 3.6×10^2 (b) 3.6×10^{-2} (c) 3.6×10^{-3} (d) 3.6×10^{-4} (e) 3.6×10^{-5}
12. If $\cos = a/b$, find $1 + \tan a$. a. b^2 / a^2 b. A^2 / b^2 c. $(A^2 + b^2) / (b^2 + a^2)$ d. $2(a^2 + b^2) / (a^2 + b^2)$ e. B/a^2
13. If the mean of five consecutive integers is 30, find the largest of the numbers a. 28 b. 30 C. 32 d. 34
14. Simplify without using tables $(\log_2 6 - \log_2 3) / (\log_2 8 - 2\log_2 1/2)$ a. $1/5$ b. $1/2$ c. $-1/2$ d. $\log_2 3 / \log_2 2$ e. $1/4$
15. If the 9th term of an A. P is five times the 5th term, find the relationship between a and d. a. $2a + d = 0$ b. $3a + 5d = 0$ c. $a + 3d = 0$ d. $A + 2d = 0$ e. $5d + a = 0$
16. A bucket is 12cm in diameter at the top, 8cm at the bottom and 4cm deep. Calculate its volume. a. $304 / \text{cm}^3$ b. 144 cm^3 c. $128 / \text{cm}^3$ d. 72 cm^3 e. 54 cm^3
17. Calculate $3310_5 - 1442_5$ a. 1313_5 b. 2131_5 c. 4303_5 d. 1103_5 e. 2213_5
18. Express the product of 0.0014 and 0.011 in standard form. (a) 1.54×10^{-2} (b) 1.54×10^{-3} (c) 1.54×10^{-4} (d) 1.54×10^{-5} (e) 1.54×10^{-6}
19. Factorize completely the expression $abx^2 + 6y - 3ax - 2yx$ (a) $(ax - 2y)(bx - 3)$ (b) $(bx + 3)(2y - ax)$ (c) $(bx + 3)(ax - 2y)$ (d) $(ax - 2y)(ax - b)$ (e) $(ax + 1)(3 - y)$
20. Find the simple interest rate percent per annul at which N1000 accumulates to N1240 in 3year. (A) 6% (b) 8% (c) 10% (d) 12% (e) 14%
21. An arc of a circle suspends on angle 70° at the centre, If the radius of the circle 6cm, calculate the area of the sector subtended by the given angle. a. 22 cm^3 b. 44 cm^3 c. 66 cm^3 d. 88 cm^3 e. 108 cm^3



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22. A final examination requires that a student answer any 4 out of 6 questions. In how many ways can this be done? (a) 15 (b) 20 (c) 30 (d) 45 (e) 360
23. Differentiate with respect to X. $6x^2 - 5x^2 + 1/3x^2$ (a) $2 + 2/3x^2$ (b) $2 + 1/6x$ (c) $2 - 2/3x^2$ (d) $2 - 1/6x$ (e) $2 + 1/3x$
24. Divide $2x^3 + 11x^2 + 17x + 6$ by $2x$ (A) $X^2 + 5x + 6$ (b) $2x^2 + 5x + 6$ (c) $2x^3 - 5x + 6$ (d) $x^2 - 5x + 6$ (e) $x - 5x + 6$
25. If $10112 + X7 = 2510$ solve for X. (a) 14 (b) 20 (c) 24 (d) 25 (e) 28

ANSWERS

1. D 2. C 3. D 4. D 5. D 6. D 7. B 8. C 9. D 10. A 11. D 12. B 13. A 14. C 15. E 16. C 17. D 18. A 19. D 20. A 21. B 22. A 23. E 24. C 25. A 26. A
26. If X is the addition of the prime numbers between 1 and 6 and Y the H.C.F of 6, 9, 15. Find the product of X and Y. (a) 27 (b) 30 (c) 33 (d) 90
27. A 5.0g of salt was weighed by Tunde as 5.1g. What is the percentage error? (a) 20 (b) 2 (c) 1 (d) 0.02
28. Two sisters, Taiwo and Kehinde, own a store. The ratio of Taiwo's share to Kehinde's is 11:9. Later, Kehinde sells $2/3$ of her share to Taiwo for 720.00. Find the value of the store. (a) 1,080.00 (b) 2,400.00 (c) 3,000.00 (d) 3,600.00
29. A basket contains green, black and blue balls in the ratio 5:2:1. If there are 10 blue balls, find the corresponding new ratio when 10 green and 10 black balls are removed from the basket (a) 1:1:1 (b) 4:2:1 (c) 5:1:1 (d) 4:1:1
30. A taxpayer is allowed $1/8$ th of his income tax fee, and pays 20% on the remainder. If he pays N490.00 tax, what is his income? (A) 560.0 (b) 2,450.00 (c) 2,800.00 (d) 3,920.00
31. Evaluate $2700000 \times 0.03 + 18000$ (a) 4.5×10^0 (b) 4.5×10^1 (c) 4.5×10^2 (d) 4.5×10^3
32. The prime factors of 2,520 are (a) 2, 9, 5 (b) 2, 9, 7 (c) 2, 3, 5, 7 (d) 2, 3, 7, 9
33. If $12e = X7$, find X where $e = 12$ (a) 20 (b) 15 (c) 14 (d) 12
34. What is the difference between 0.007685 correct to three significant figures and 0.007685 correct to four places of decimal? (a) 10^5 (b) 7×10^{-4} (c) 8×10^{-5} (d) 10^{-5}
35. Oke deposited N800.00 in the bank at the rate of $12\frac{1}{2}\%$ simple interest. After some time the total amount was one and half times the principal. For how many years was the money left in the bank? (a) 2 (b) 4 (c) $5\frac{1}{3}$ (d) 8
36. If the surface area of a sphere is increased by 44%, find the percentage increase in its diameter. (a) 44 (b) 30 (c) 22 (d) 20
37. In a small village of 500 people, 350 speak the local language while 200 speak pidgin English. What percentage of the population speaks both? (A) 30% (b) 10% (c) 50% (d) 14%



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38. Factorize $4a^2 - 12ab - C^2 + 9b^2$ (a) $4a(a-3b)+(3b-c)^2$ (b) $(2a+3b-c)(2a+3b+c)$ (c) $(2a-3b-c)(2a-3b+c)$ (d) $4a(a-3b)+(3b+c)^2$
39. What are K and L respectively if $\frac{1}{2}(3y-4x)^2 = (8x^2 + Kxy + Ly^2)$ (a) -12,9/2 (b) -6,9 (c) 6,9 (d) 12,9/2
40. If M represents the median and D is the mode of the measurements 5, 9, 3 and 5, the (M,D) is (a) (6,5) (b) (5,8) (c) (5,7) (d) (5,5) (e) (7,5)
41. Simplify 2134×234 (a) 103 114 (b) 103214 (c) 122314 (d) 13211
42. In a class of 40 students, 32 offer Mathematics, 24 offer physics and 4 offer neither mathematics nor physics. How many offer both mathematics and physics? (a) 20 (b) 16 (c) 8 (d) 4
43. A woman buys 270 oranges for N 1,800.00 and sells at 5 for N40.00. What is her profit? (A) 2,160.00 (b) 1,620.00 (c) 630.00 (d) 360.00
44. Find the simple interest rate percent per annum at which N1000 accumulates to N1240 in 3 years. (a) 6% (b) 8% (c) 10% (d) 12% (e) 14%
45. The length a person can jump is inversely proportional to his weight. If a 20kg person can jump 1.5m, find the constant of proportionality. (a) 60 (b) 30 (c) 20 (d) 15
46. Find the values of X and Y respectively if $3x - 5y + 5 = 0$ and $4x - 7y + 8 = 0$ (a) 0,4 (b) -4,-5 (c) 4,5 (d) 0.5,4
47. Divide $6x^2 - 13x + 5$ by $2x - 1$ (a) $3x+5$ (b) $3x-5$ (c) $5x-3$ (d) $5x+3$
48. A polynomial in X whose zeros are -2,-1 and 3 is (A) x^3-7x+6 (b) x^3+7x-6 (c) x^3+7x+6 (d) x^3-7x-6
49. The time taken to do a piece of work is inversely proportional to the number of men employed. If it takes 30 men to do a piece of work in 6 days, how many men are required to do the work in 4 days? (A) 20 (b) 35 (c) 5 (d) 60

ANSWERS

25.B 26.B 27.B 28.A 29.B 30.D 31.B 32.A 33.C 34.C 35.D 36.A 37.B 38.C 39.C 40.D 41.D 42.C 43.A 44.B 45.A 46.B 47.C 48. 49. D 50. C

1. A student measures a piece of rope and found that it was 1.26 m long. If the actual length of the rope was 1.25 m, what was the percentage error of the measurement? A. 0.40% B. 0.0 1% C. 0.25% D. 0.80%.
2. At what rate will the interest on N400 increase to N 24 in 3 years reckoning in simple interest? A. 4% B. 2% C. 3% D. 5%.



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3. Given that $\log 2 = 0.3010$, $\log 7 = 0.8451$. Evaluate $\log 112$. A. 2.5441 B. 2.049 1 C. 2. 1461 D. 3.1461.
4. Express the product of 0.21 and 0.34 in standard form A. 7.14×10^{-3} B. 7.14×10^{-1} C. 7.14×10^{-2} D. 7.14×10^{-4}
5. In a survey of 50 newspaper readers, 40 read Champion and 30 read Guardian, how many read both papers? A. 15 B. 5 C. 10 D. 20.
6. If $9x^2 + 6xy + 4y^2$ is a factor of $27x^3 - 8y^3$, find the other factor. A. $3x - 2y$ B. $2y - 3x$ C. $2y + 3x$ D. $3x + 2y$
7. Solve for x and y if $x - y = 2$ and $x^2 - y^2 = 8$ A. (1, 3) B. (3, 1) C. (-1, 3) D. (-3, 1).
8. If y varies directly as the square root of x and $y = 3$ when $x = 16$. Calculate y when $x = 64$. A. 5 B. 12 C. 6 D. 3
9. If x is inversely proportional to y and $x = 2\frac{1}{4}$ when $y = 2$, find x if $y = 4$. A. $2\frac{1}{4}$ B. 5 C. 4 D. $1\frac{1}{4}$
10. Find the sum to infinity of the following series. $0.5 + 0.05 + 0.005 + 0.0005 \dots$ A. $\frac{5}{9}$ B. $\frac{5}{7}$ C. $\frac{5}{11}$ D. $\frac{5}{11}$
11. The 3rd term of an arithmetic progression is -9 and the 7th term is -29. Find the 10th term of the progression. A. 44 B. -165 C. -44 D. 165.
12. If the length of a square is increased by 20% while its width is decreased by 20% to form a rectangle, what is the ratio of the area of the rectangle to the area of the square? A. 6:5 B. 25:24 C. 5:6 D. 24:25
13. Two brothers invested a total sum of N5, 000.00 on a farm project. The farm yield was sold for N15,000.00 at the end of the season. If the profit was shared in the ratio 2:3, what is the difference in the amount of profit received by the brothers? A. N2, 000.00 B. N4, 000.00 C. N6, 000.00 D. N10,000.00.
14. Peter's weekly wages are N20.00 for the first 20 weeks and N36.00 for the next 24 weeks. Find his average weekly wage for the remaining 8 weeks of the year, if his average weekly wage for the whole year is N30.00. A. N37.00 B. N35.00 C. N30.00 D. N25.00
15. A man invests a sum of money at 4% per annum simple interest. After 3 years the principal amounts to N7, 000.00. Find the sum invested. A. N7, 840.00 B. N6, 250.00 C. N6,160.00 D. N5,833.33.
16. By selling 20 oranges for N1.35 a trader makes a profit of 8%. What is his percentage gain or loss if he sells the same 20 oranges for N1.10? A. 8% B. 10% C. 12% D. 15%.
17. Four boys and ten girls can cut a field in 5 hours. If the boys work at $\frac{5}{4}$ hrs rate of which the girls work, how many boys will be needed to cut the field in 3 hours? A. 180 B. 60 C. 25 D. 20



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18. The factors $6x-5-x^2$ are (a) $(x+3)$, $(x+2)$ (b) $(x-5)$, $(x-1)$ (c) $-(x+5)$, $(X+1)$ (d) $(x-5)$, $(1-x)$ (e) $(x+5)$, $(1x)$
19. Seven years ago, the age of a father was three times that of his son, but in six years time the age of the son will be half of the father. Representing the ages of the father and son by X and Y, respectively, the two equations relating X and Y are; (a) $3Y-X=0$; $2Y-X=0$ (b) $3Y-X=14$; $X-2Y=6$ (c) $3Y-X=7$; $X-2Y=6$ (d) $3Y-X=14$; $Y-2X=6$ (e) $X+3Y=7$; $X=2Y=12$
20. If M represents the median and D is the mode of the measurements 5, 9, 3 and 5, the (M,D) is (a) (6,5) (b) (5,8) (c) (5,7) (d) (5,5) (e) (7,5)
21. The value of $(0.03)^2 - (0.02)^2$ is (a) 0.019 (b) 0.0019 (c) 0.00019 (d) 0.000019 (e) 0.000035
22. If $263+441=714$, what number base has been used? (a) 12 (b) 11 (c) 10 (d) 9 (e) 8
23. What is the probability of having an odd number in a single toss of a fair die? A. $1/6$ (b) $1/3$ (c) $1/2$ (d) $2/3$ (e) $5/6$
24. A sales girl gave a change of N1.15 to a customer instead of N1.25. Calculate the Percentage error (a) 10% (b) 8.7% (c) 8.0% (d) 2.4% (e) 0.10%
- 25.

Score	0	1	2	3	4	5
Frequency	2	3	4	2	7	2

Find the mean score in the frequency table above showing the performance a group of student in a test.

- (A) 1.75 (b) 2 (c) 2.5 (d) 2.75 (e) 3

ANSWERS

1. D 2. B 3. B 4. C 5. D 6. A 7. B 8. C 9. D 10. A 11. C 12. 13. A 14. A 15. B 16. C 17. D 18. C 19. D 20. D 21. D 22. D 23. C 24. C 25. D

GENERAL NOTE

Benefits of Studying Past Exam Papers

Studying past papers is a valuable part of exam preparation and help keep revision **focused** on important themes whilst practicing exam style questions.

Past exam papers are one of the most helpful tools available to prepare for both internal and external examinations as they provides students with **practical insight** into how the forthcoming exam paper is likely to look and the **key themes** or **subject areas** most likely to be covered. Before starting revision for a subject it is always worth taking time to seek out past papers.

Please note that this is a post UTME past question and not post UTME Sure Questions and therefore it is meant to be practice and also guide you in your revision. The number of questions here may not be the exact number of questions you will be given in your case. Also the time you will be given may vary from the one indicated in this text. Therefore, Classic Educational Consults will not be responsible for any form of Examination Malpractice or Examination Misconduct.

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